

UNIVERSITY OF TRIER
Overview of Used Generative AI Tools and Digital Aids
Forschungspraktikum Wirtschaftsinformatik – Sommersemester 2026

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In accordance with the guidelines of the *Forschungspraktikum* and the official regulations of Trier University, we hereby declare the comprehensive usage of generative artificial intelligence (GenAI) models and digital tools during this research project.

1. Utilized GenAI Models and Platforms:

- **JetBrains Junie** (integrating Google Gemini 3.6 Flash, 3.7 Flash, and 3.8 Flash)
- **OpenAI ChatGPT** (utilizing ChatGPT-5.6 Sol and ChatGPT-6 Astra)

2. Literature Comprehension and Background Research: Literature research was conducted manually using literature databases and a structured snowballing approach without automated AI search agents. GenAI tools were utilized strictly to aid in explaining and comprehending complex domain concepts, mathematical formulations (e.g., RoA calculations), and underlying methodology from source literature. All conceptual insights gained were independently verified against the original publications.

3. Critical Reflection on Research Questions and Methodology: Following our independent initial formulation of the research questions and experimental methodology, GenAI models were employed to critically review and stress-test our design. Specifically, the tools were prompted to evaluate the precision, scope, and time feasibility of our research questions regarding infrastructure cascading failures. Suggestions were critically evaluated, and final adjustments were made autonomously.

4. Software Development and Code Implementation: During software development, GenAI served extensively as an interactive implementation assistant embedded within the IDE. We independently designed the overall software architecture, FSM logic, network routing algorithms, data models, and pipeline execution flow. Prior to coding, we formulated explicit execution plans, functional specifications, expected outputs, and testing strategies for each component. GenAI was then instructed with these highly specific technical directives to write discrete functions and classes, draft geospatial data preprocessing scripts, build visualization components, translate our predefined concepts into code, and assist in navigating legacy codebases. Every line of AI-assisted code was manually reviewed, understood, refactored, and tested against our predefined validation criteria to ensure complete execution correctness.

5. Academic Writing, Style Refinement, and Proofreading: The entire manuscript was initially written and drafted independently by the authors. GenAI was subsequently used exclusively as a post-writing editing aid to correct grammar, refine vocabulary, improve sentence flow, and polish the academic tone **throughout the entirety of the paper**. Text suggestions were critically evaluated on an individual, sentence-by-sentence basis before integration. No scientific content, statistical interpretations, or manuscript sections were autonomously generated by AI.

Trier, 30.09.2026
Place, Date


Signature (Joé Welsch)

Trier, 30.09.2026
Place, Date


Signature (Van Bao Nguyen)